Program : **Diploma in Automation and Robotics**	
Course Code : **6339C**	Course Title: **Mechatronics and Automation Lab**
Semester : **6**	Credits: **1.5**
Course Category: **Engineering Science**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- Measuring of physical quantity (i) Velocity (ii) direction and (iii) force of single and double acting actuators also the operation of signal conditioning circuits.
- Applying a suitable sensor and image processing technique for Mechatronics systems.
- Design appropriate circuits to automate and control the hydraulic, pneumatic and electric actuators.
- Apply PLC, PID and 8085 microcontrollers as a control unit Mechatronics system.

General Instructions:

- Students have to write the date, aim, Software & Hardware requirements for that experiment in the observation book.
- Students have to listen and understand the experiment explained by the faculty and note down the important points in the observation book.
- Students need to write procedure/algorithm in the observation book.
- Analyze and develop /implement the logic of the program by the student in respective platform
- After approval of logic of the experiment by the faculty then the experiment has to be executed on the system.
- After successful execution the results are to be shown to the faculty and noted the same in the observation book.
- Students need to attend the Viva-Voce on that experiment and write the same in the observation book.
- Update the completed experiment in the record and submit to the concerned faculty in-charge.

Tools and Equipment:

1. Basic Pneumatic Trainer Kit with manual and electrical controls/PLC Control - 1 each
2. Basic Hydraulic Trainer Kit - 1 No.
3. Hydraulics and Pneumatics Systems Simulation Software's - 10 sets.
4. 8051 - Microcontroller kit with stepper motor and drive circuit - 2 sets
5. Simulation Software's and Sensors to measure Pressure, Flow rate, direction, speed, velocity and force. -2 sets

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Measuring of physical quantity (i) Velocity (ii) direction and (iii) force of single and double acting actuators also the operation of signal conditioning circuits.	8	Applying
CO2	Applying a suitable sensor and image processing technique for Mechatronics systems.	8	Applying
CO3	Design appropriate circuits to automate and control the hydraulic, pneumatic and electric actuators	8	Applying
CO4	Apply PLC, PID and 8085 microcontrollers as a control unit Mechatronics system.	8	Applying
CO5	Developing a model of pneumatic and hydraulic circuits by using simulation software.	10	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3						
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Measuring of physical quantity (i) Velocity (ii) direction and (iii) force of single and double acting actuators also the operation of signal conditioning circuits.**		

M1.01	Introduction of mechatronics system with velocity sensors.	2	Applying
M1.02	Introduction of mechatronics system with direction sensors.	2	Applying
M1.03	Introduction of mechatronics system with single and double acting actuators.	2	Applying
CO2	**Applying a suitable sensor and image processing technique for Mechatronics systems.**		
M2.01	Speed Control of AC & DC drives.	2	Applying
M2.02	Computerized data logging system with control for process variables like pressure flow and temperature.	3	Applying
M2.03	Servo controller interfacing for DC motor	3	Applying
	Lab Exam I	1.5	
CO3	**Design appropriate circuits to automate and control the hydraulic, pneumatic and electric actuators.**		
M3.01	Design of circuits with logic sequence using Electro pneumatic trainer kits.	2	Applying
M3.02	Simulation of basic Hydraulic, Pneumatic and Electric circuits using software.	3	Applying
M3.03	Modeling and analysis of basic electrical, hydraulic and pneumatic systems Using appropriate softwares.	3	Applying
CO4	**Apply PLC, PID and 8085 microcontrollers as a control unit Mechatronics system.**		
M4.01	PID controller interfacing	4	Applying
M4.02	Circuits with multiple cylinder sequences in Electro pneumatic using PLC	4	Applying
CO5	**Developing a model of pneumatic and hydraulic circuits by using simulation software.**		
M5.01	Simulation of basic Hydraulic, Pneumatic and Electric circuits using software.	5	Applying

M5.02	Modeling and analysis of basic electrical, hydraulic and pneumatic systems Using appropriate softwares.	5	Applying
M5.03	Lab Exam II	1.5	